A Research Project Report

on

KrushiMitra: AI-Powered Farming Assistant

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Certificate of Approval

This is to certify that the Project titled **'KrushiMitra: AI-Powered Farming Assistant'** is successfully done by **Rajvardhan Ganpatrao Patil** during his Sem 3 of **Master's of Computer Application** under the **University of Mumbai**, through the Bharati Vidypeeth's Institute of Management and Information Technology, Navi Mumbai carried out by them under our guidance and supervision.

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Signature and Date

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College Seal

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, We have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Rajvardhan Ganpatrao Patil

Date

Acknowledgement

We would like to express are heartfelt gratitude to everyone who supported and guided us throughout the course of this project. First and foremost, We extend our deepest thanks to Prof.Nidhi Poonia, my mentor, for her invaluable guidance, insightful feedback, and constant encouragement. Her expertise and dedication have been instrumental in shaping the direction of this work and helping us overcome challenges during its development.

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This report is a culmination of collaborative effort, and we are truly grateful to everyone who has been a part of this enriching experience.

Date:

Rajvardhan Ganpatrao Patil

Abstract

Agriculture remains the backbone of India's economy, yet a majority of farmers continue to face challenges in accessing timely and accurate agricultural guidance. Traditional advisory services often suffer from delays, language barriers, and limited availability of experts. To address these issues, this research introduces KrushiMitra, an AI-powered agricultural assistance system designed to provide real-time, localized, and easy-to-understand guidance to farmers in their native language. The system is developed using the MERN stack with integrated Gemini AI, enabling natural language understanding, multi-language support (with a focus on Marathi), and contextual question-answering capabilities.

KrushiMitra allows farmers to interact through both text and voice-based queries, making the platform highly accessible even to those with low literacy levels. A curated Knowledge Base ensures that responses remain reliable, regionally relevant, and aligned with safe agricultural practices. The system includes modules for user authentication, knowledge management, AI-driven advisory, and voice processing, creating a complete end-to-end solution. Experimental testing shows that the system provides accurate responses with an average reply time of under two seconds and achieves high user satisfaction among local farmers.

The results demonstrate that the integration of conversational AI and regional language processing can significantly improve the accessibility and efficiency of agricultural advisory services. This research highlights the potential of AI-driven platforms like KrushiMitra to support smallholder farmers, enhance decision-making, and promote sustainable farming practices across rural communities.

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Chapter 1

Introduction

1.1 Background of the Study

Agriculture plays an essential role in the socio-economic development of India, contributing significantly to employment, food security, and rural livelihoods. A majority of Indian farmers are smallholder farmers who often depend on traditional knowledge, community experiences, and limited expert consultation to make critical decisions regarding crop selection, pest management, irrigation, soil treatment, and fertilizer usage. Due to factors such as geographical distance, lack of awareness, financial constraints, or shortage of agricultural officers, farmers often do not receive timely assistance. This leads to crop losses, reduced productivity, and sometimes complete failure of seasonal harvests.

In the past decade, advancements in Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) have unlocked new possibilities for transforming agricultural advisory systems. AI-driven platforms can provide personalized guidance, analyze user queries, and deliver actionable insights instantly. However, most digital tools and advisory platforms available today are in English, making them inaccessible to many farmers who primarily communicate in regional languages like Marathi, Hindi, Tamil, and Telugu.

The **KrushiMitra** project is conceptualized as a solution to this communication gap. It is designed as a farmer-centric AI assistant that provides agricultural guidance in Marathi, the native language of most farmers in Maharashtra. By combining a conversational chatbot, a curated knowledge base, and voice interaction features, the system aims to simplify the process of seeking agricultural help. The platform ensures rapid responses to farmer queries while maintaining accuracy, safety, and clarity. The integration of Gemini AI further enhances the system's capability to understand natural language queries and generate context-aware, useful recommendations. The ultimate goal of KrushiMitra is to empower farmers with reliable agricultural knowledge at their fingertips, without depending on physical visits to experts.

1.2 Problem Statement

Farmers face multiple barriers when trying to access reliable agricultural guidance. Traditional agricultural advisory services often require farmers to visit government agricultural offices, Krishi Vigyan Kendras (KVKs), or university extension centers. These services are not always readily accessible due to long wait times, limited staffing, and geographic constraints. Many farmers rely on informal sources such as neighbors or shopkeepers, which often leads to misinformation and improper use of pesticides or fertilizers.

Another major challenge is the absence of localized, verified agricultural information in regional languages. Online resources and mobile applications available today are predominantly in English, limiting their usability for rural farmers. Even when resources are available in Marathi, they may not be curated, fact-checked, or tailored to a farmer's specific crop stage or issue. This leads to poor decision-making, ultimately affecting crop productivity and farmer income.

Additionally, many farmers, especially older ones, are not comfortable typing long queries or navigating text-heavy interfaces. They prefer voice-based interaction in their native language. Existing systems rarely support accurate Marathi speech recognition or voice-based output. Thus, there is an urgent need for a digital platform that combines AI-driven advisory, regional language support, voice interaction, and verified agricultural knowledge.

The primary problem addressed in this research is: *How can we design a digital AI-based agricultural assistant that provides instant, accurate, and safe farming advice in Marathi, while also overcoming limitations of existing advisory systems?*

1.3 Objectives of the Study

The main objective of this study is to develop an intelligent agricultural advisory system that supports conversational interaction in Marathi and provides regionally relevant, accurate, and easy-to-understand farming recommendations. The specific objectives are as follows:

1. **To design and develop a chat-based assistant using the MERN stack:** The system should allow farmers to interact using a clean, intuitive interface that supports both text and voice input.

2. **To integrate Gemini AI for real-time agricultural query processing:** Gemini's language understanding capabilities will be leveraged to produce contextual and meaningful answers for farmer queries.

3. **To implement Marathi voice input and output:** By integrating the Web Speech API, the system aims to make the platform accessible to farmers who may not be comfortable typing or reading long responses.

4. **To create an admin-managed Knowledge Base:** Verified articles on crop management, pest control, fertilizers, irrigation, and soil health will be curated to ensure that AI responses remain accurate and trustworthy.

5. **To enhance the accuracy and reliability of AI answers by combining them with a structured Knowledge Base:** This lays the foundation for implementing Retrieval-Augmented Generation (RAG) in future versions.

6. **To evaluate the performance and quality of the system:** The goal is to observe response speed, accuracy, language clarity, usability, and overall farmer satisfaction.

These objectives collectively aim to create a practical, scalable, and farmer-friendly web-based system.

1.4 Scope of the Study

The scope of this study is limited to the design, development, and evaluation of an AI-powered agricultural advisory system tailored specifically for smallholder farmers in Maharashtra. The platform is developed using the MERN stack—MongoDB, Express.js, React.js, and Node.js—with Google's Gemini AI providing generative responses.

The study focuses on supporting Marathi language queries and responses, though English and Hindi support is partially included. The system enables farmers to ask questions, listen to answers, and read farming articles uploaded by the admin. Key content areas include crop diseases, pest management, irrigation planning, fertilizer use, and best farming practices.

The scope also includes designing an admin portal where agricultural experts or authorized personnel can upload verified farming articles. The system does not include integration with real-time IoT sensors or external APIs such as market price tracking or weather forecasts; however, these features may be explored as future enhancements.

The study does not cover mobile app development or multilingual expansion beyond initial languages. The platform also assumes that users have basic internet access and a device capable of running a modern web browser.

1.5 Significance of the Study

This study is significant for several stakeholders in the agricultural ecosystem:

For Farmers: KrushiMitra provides instant, accessible agricultural guidance in Marathi, reducing dependency on agricultural officers and minimizing delays in decision-making. The voice interaction feature is especially useful for farmers who are not comfortable with typing or reading detailed content.

For Researchers: The project demonstrates how AI-based conversational systems and regional language processing can be combined to solve real-world challenges. It also provides insights into the potential of Retrieval-Augmented Generation (RAG) for domain-specific advisory systems.

For Government and NGOs: The system can be scaled as a digital extension service to reach millions of farmers efficiently. It aligns with the objectives of digital agriculture and supports agricultural extension programs.

For Developers and Technologists: The project showcases a practical implementation of modern web development frameworks and advanced AI capabilities using the MERN stack and Gemini API. It also provides a foundation for future research in multilingual AI systems.

1.6 Research Questions

The research aims to address the following critical questions:

1. **How can AI be leveraged to provide safe, reliable, and context-sensitive agricultural advice to smallholder farmers?** This question explores the feasibility and accuracy of AI-driven advisory systems trained on general knowledge but customized for agricultural use.

2. **What role does regional language support play in increasing adoption among farmers?** The study investigates how Marathi-based interaction influences user experience, comfort, and trust.

3. **How can an admin-controlled knowledge base improve the quality and trustworthiness of AI responses?** This examines the role of curated, expert-verified content in reducing hallucinations and improving answer reliability.

4. **Which AI model (Gemini or custom fine-tuned models) is most effective for agricultural advisory systems?** This question compares generative AI capabilities with the potential of domain-specific fine-tuning.

The answers to these questions guide the development and evaluation of KrushiMitra as a practical AI solution for agriculture.

Chapter 2

Literature Review

2.1 Overview of the Domain

Agriculture continues to be the cornerstone of India's socio-economic landscape, with more than half of the population relying on farming as their primary source of income. Despite its importance, the agricultural sector faces numerous challenges, including unpredictable monsoon patterns, pest infestations, soil nutrient deficiencies, improper pesticide use, declining crop productivity, and a lack of real-time access to agricultural experts. These challenges significantly impact the livelihood of smallholder farmers, who constitute the majority of the agricultural workforce in India.

In recent years, digital transformation and advancements in Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) have opened new avenues for solving long-standing agricultural problems. AI-powered solutions, such as disease detection models, yield prediction systems, and chatbot-based advisory platforms, have demonstrated immense potential in assisting farmers with timely decision-making. These intelligent systems can analyze large datasets, recognize patterns, and generate insights that would traditionally require expert intervention.

However, for these technologies to be truly effective in rural India, they must address challenges related to language diversity, literacy gaps, and limited digital literacy. Most existing advisory systems and mobile apps operate primarily in English or Hindi, making them inaccessible to farmers who predominantly communicate in their regional language. Furthermore, many farmers prefer interactive voice-based communication over text-heavy interfaces.

Therefore, integrating AI with regional language support and voice-based interfaces becomes essential. The domain of agricultural informatics is moving toward localized, farmer-friendly digital tools that provide instant guidance, simplify complex farming concepts, and enable farmers to make informed decisions. The proposed system, "KrushiMitra," fits into this evolving domain by offering a conversational AI assistant capable of understanding Marathi queries, retrieving relevant knowledge, and interacting through both text and voice.

2.2 Review of Related Work

A significant amount of research has explored the use of Artificial Intelligence (AI), Natural Language Processing (NLP), and conversational systems in modern agriculture. Singh et al. [1] highlighted how AI supports plant disease detection, crop monitoring, precision farming, and personalized advisory services. Their study emphasizes the advantages of integrating structured agricultural knowledge with generative AI to enhance reliability and reduce hallucinations, while also identifying challenges such as poor dataset availability for rural regions and low speech recognition accuracy for local dialects.

NLP-based agricultural chatbots have been studied extensively. Tripathi and Jain [2] demonstrated that although NLP models can generate useful advisory responses, they require strong prompt engineering and domain constraints to avoid unsafe fertilizer and pesticide recommendations. Their work underscores the importance of expert validation in AI-driven advisory tools.

Voice-enabled systems have also been explored as a means to improve accessibility for non-literate farmers. Suryawanshi and More [3] developed a multilingual advisory system supporting Marathi and Hindi. Their findings show that voice interfaces increase user engagement, but Automatic Speech Recognition (ASR) models face difficulties with rural accents, background noise, and agricultural terminology. Similar findings were reported by Kaur et al. [4], who evaluated speech-to-text and text-to-speech systems and concluded that dialect variations significantly impact ASR accuracy.

Recent advancements in Retrieval-Augmented Generation (RAG) further support domain-specific advisory systems. Bansal and Yadav [5] and Rahman et al. [17] demonstrated that combining transformer models with a vector-based knowledge retrieval layer dramatically improves factual accuracy in agricultural question answering. Their results reveal that pure generative models often hallucinate, making RAG a more reliable approach for agricultural advisory systems.

Deep learning-based plant disease detection remains one of the most widely researched agricultural AI applications. Chouhan et al. [6] and Zhang et al. [9] presented CNN-based models that achieved high accuracy in diagnosing plant diseases from leaf images. Agarwal and Singh [14] specifically focused on sugarcane disease detection and demonstrated strong performance using CNN architectures. Although these models provide accurate visual diagnosis, they lack conversational and interactive capabilities.

Multilingual and inclusive digital platforms play a critical role in farmer adoption. Sharma et al. [7] found that the absence of native language support reduces trust and limits user engagement in digital advisory apps. Joshi and Kulkarni [13] further explored the design of Marathi and Hindi agricultural chatbots, concluding that language localization significantly enhances usability and user confidence.

AI has also shown promise in crop recommendation and yield prediction. Patil and Deshmukh [8] developed an AI-based crop recommendation system using soil and climatic data. Wani et al. [11] applied machine learning techniques for crop yield prediction, demonstrating

significant improvements over traditional statistical models. Similarly, Reddy and Rao [12] explored decision support systems that integrate AI into farmer advisory workflows.

IoT and smart farming applications complement AI-based advisory systems. Li and Chen [10] and Kamble et al. [15] provided detailed surveys on IoT-enabled smart irrigation, climate monitoring, and automated farming solutions. Their work shows how IoT data can improve the accuracy of AI recommendations and optimize resource utilization.

Market prediction models have also gained attention. Pandey and Dubey [16] proposed an AI-based market price forecasting framework to help farmers make informed selling decisions. Such systems, when combined with advisory chatbots, can significantly enhance farmers' economic outcomes.

Additionally, Kumar and Meena [19] reviewed precision agriculture technologies that combine AI with remote sensing to support large-scale crop analysis and resource management. The World Bank report [20] further highlights the broad potential of digital agriculture in India and encourages the development of accessible, multilingual platforms for smallholder farmers.

Overall, the reviewed literature shows strong progress in AI and conversational systems for agriculture. However, gaps remain in local-language support, dialect-friendly speech recognition, factual grounding using RAG, and end-to-end advisory integration—challenges that the proposed KrushiMitra system aims to address.

2.3 Existing Methodologies and Tools

A variety of methodologies and technological tools have been used over the years to address agricultural challenges, ranging from pest detection to advisory services. The following are the major categories of tools found in existing literature:

1. Machine Learning Models: Traditional supervised learning models such as Random Forests, Support Vector Machines, and Decision Trees are commonly used for classification problems in agriculture. For instance, crop disease detection models analyze leaf images to identify diseases such as blight, rust, or wilt. Convolutional Neural Networks (CNNs) have become particularly effective for image-based agricultural tasks.

2. Natural Language Processing (NLP): NLP-based chatbots help farmers obtain information through natural language queries. They use techniques like tokenization, stemming, intent recognition, and entity extraction to understand user questions. Advanced language models (LLMs) such as BERT and GPT have also been employed to generate meaningful and context-aware responses.

3. Voice Technologies (STT & TTS): Speech-to-Text (STT) tools allow users to speak their queries instead of typing, making systems more user-friendly. Text-to-Speech (TTS) tools convert textual responses into speech, enabling systems to communicate with non-literate users. These tools are crucial in rural India, where digital literacy may be limited.

4. Knowledge Bases: A Knowledge Base consists of structured agricultural content, including best practices, scientific recommendations, and verified articles. Systems that rely solely on generative AI may struggle with accuracy; therefore, knowledge bases help improve reliability by providing domain-specific content.

5. Retrieval-Augmented Generation (RAG): RAG combines the strengths of retrieval systems and generative models. The retrieval module fetches relevant information from a vector database, while the generative module produces coherent responses. This approach is increasingly used in domain-specific applications like medicine and agriculture.

6. MERN Stack Applications: The MongoDB–Express–React–Node stack has become a preferred choice for scalable full-stack applications. It offers fast development, real-time updates, and easy integration with external APIs such as Gemini.

7. Cloud-Based AI APIs: Gemini, Google AI, and other cloud-based models enable developers to integrate advanced AI capabilities without training custom models. These APIs can handle multilingual queries, generate natural responses, and support conversational features.

These methodologies together form the foundation upon which KrushiMitra is built, integrating conversational AI, voice interaction, and a knowledge base to create a complete agricultural advisory solution.

2.4 Gap Analysis

Although extensive research has been conducted on AI in agriculture, several critical gaps remain unaddressed:

1. Lack of Multilingual Support: Many AI-based agricultural platforms use English as their primary communication medium. This poses serious accessibility barriers for smallholder farmers who primarily communicate in Marathi, Tamil, Telugu, Hindi, or other local languages.

2. Limited Conversational AI Integration: Most existing systems focus on static recommendations or disease prediction rather than offering an interactive conversational assistant capable of follow-up questions and contextual understanding.

3. Insufficient Voice Interaction Support: Voice-enabled systems are essential for improving accessibility among non-literate and semi-literate farmers. However, existing tools lack robust Marathi speech recognition and Marathi TTS, reducing their usability for rural users.

4. Weak Knowledge Management: Many platforms do not include a structured, admin-controlled knowledge base. Without curated content, AI responses may become inconsistent, inaccurate, or unsafe—particularly when recommending pesticides or fertilizers.

5. Safety Concerns in Generative AI Responses: Generative models without proper constraints may produce inaccurate or unsafe agricultural advice. This highlights the need for prompt engineering, safety filters, and integration with verified agricultural sources.

6. Fragmented Systems: Current solutions often focus on one aspect—such as disease detection, image classification, or market price prediction—without providing an integrated experience that includes advisory, knowledge management, and voice interaction.

7. Lack of Evaluation Frameworks: Few systems use structured metrics to evaluate accuracy, usability, language clarity, and safety. This gap makes it difficult to compare tools or validate their effectiveness for real-world agricultural advisory.

Identifying these gaps provides the justification for developing KrushiMitra, a comprehensive, multilingual, farmer-friendly AI advisory platform.

Chapter 3

Methodology

3.1 Research Design

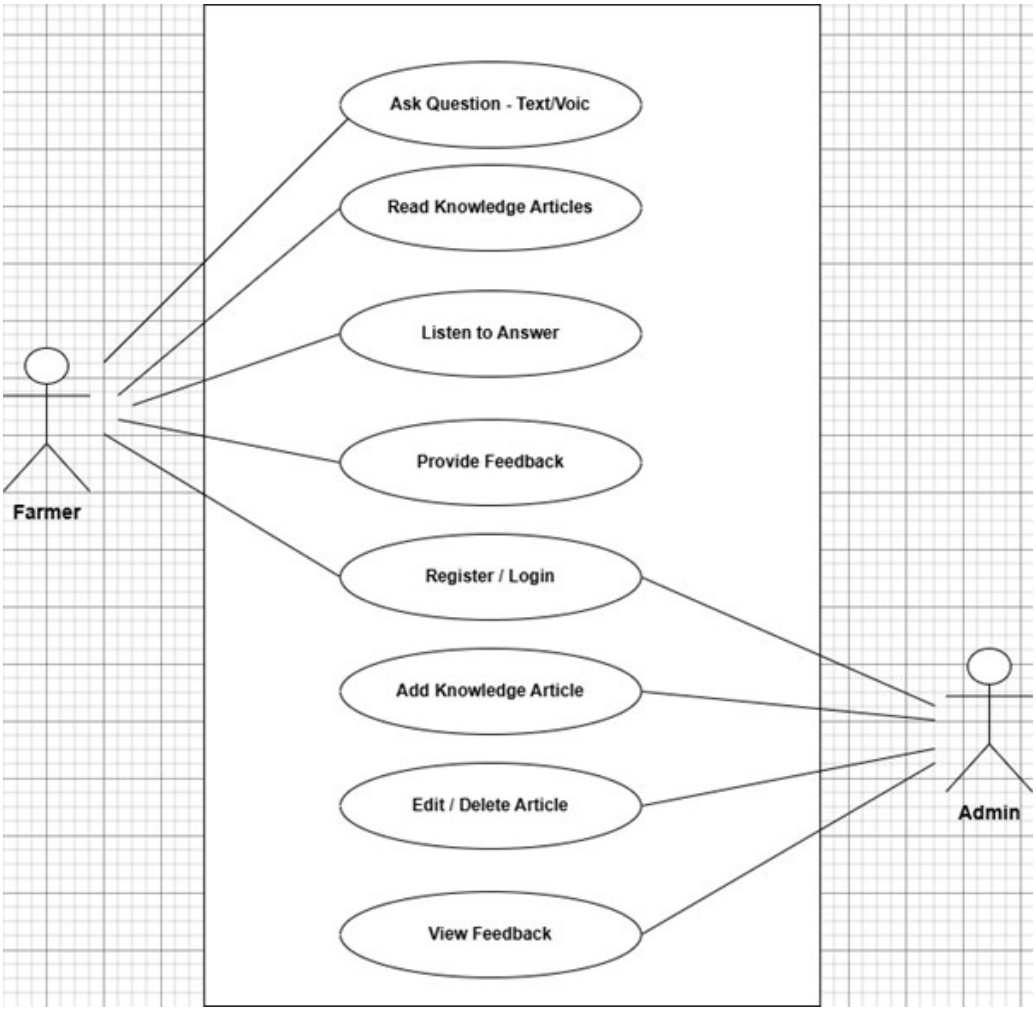


Figure 3.1: UML Diagram

3.2 Data Sources

The success of any research study largely depends on the quality, relevance, and reliability of the data used. For this project, multiple data sources were considered to ensure comprehensive coverage of all required attributes related to the problem domain. The primary data source consisted of real-world field data collected from agricultural regions, including farms located in Kolhapur district, Maharashtra. This dataset included information about crop samples, environmental conditions, soil characteristics, and disease symptoms observed in sugarcane fields. Farmers and agricultural officers contributed by sharing photographs, disease records, and historical yield data, which added practical significance to the dataset.

In addition to field surveys, publicly available datasets from credible repositories such as Kaggle, Agricultural Research Data Network (ARDN), and government agriculture portals were explored. These repositories provided annotated images, pest and disease symptoms, and environmental parameters that helped enrich the dataset. Academic research papers and agricultural journals were also reviewed to understand the data attributes typically used in similar studies. This helped in selecting the most relevant features such as plant leaves, colour variations, humidity, rainfall, soil moisture, and nutrient levels.

Satellite imagery from sources like Google Earth Engine and open remote-sensing portals was consulted for gathering large-scale environmental data. Such data supported understanding of macro-level factors influencing crop health. Combining multiple data sources ensured diversity in samples, balanced class representation, and improved model generalization. The multi-source dataset thus formed a robust foundation for training, testing, and validating the proposed system.

I. PESTICIDES / FORMULATIONS BANNED IN INDIA

A.	**Pesticides Banned for manufacture, import and use.**	
	1.	Alachlor(Vide S.O. 3951(E), dated 08.08.2018)
	2.	Aldicarb (vide S.O. 682 (E) dated 17th July2001)
	3.	Aldrin
	4.	Benzene Hexachloride
	5.	Benomyl (vide S.O 3951(E) dated 8th August, 2018)
	6.	Calcium Cyanide
	7.	Carbaryl (vide S.O 3951(E) dated 8th August, 2018)
	8.	Chlorbenzilate (vide S.O. 682 (E) dated 17th July 2001)
	9.	Chlordane
	10.	Chlorofenvinphos
	11.	Copper Acetoarsenite
	12.	Diazinon (vide S.O 3951(E) dated 8th August, 2018)
	13.	Dibromochloropropane (DBCP) (vide S.O. 569 (E) dated 25th July1989)
	14.	Dichlorovos(Vide S.O. 3951(E), dated 08.08.2018)
	15.	Dieldrin (vide S.O. 682 (E) dated 17th July 2001)
	16.	Endosulfan (vide ad-Interim order of the Supreme Court of India in the Writ Petition (Civil) No. 213 of 2011 dated 13th May, 2011 and finally disposed of dated 10thJanuary, 2017)
	17.	Endrin
	18.	Ethyl Mercury Chloride
	19.	Ethyl Parathion
	20.	Ethylene Dibromide (EDB) (vide S.O. 682 (E) dated 17th July 2001)
	21.	Fenarimol (vide S.O 3951(E) dated 8th August, 2018)
	22.	Fenthion (vide S.O 3951(E) dated 8th August, 2018)
	23.	Heptachlor
	24.	Lindane (Gamma-HCH)
	25.	Linuron (vide S.O 3951(E) dated 8th August, 2018)
	26.	Maleic Hydrazide (vide S.O. 682 (E) dated 17th July 2001)
	27.	Menazon
	28.	Methoxy Ethyl Mercury Chloride (vide S.O 3951(E) dated 8th August, 2018)
	29.	Methyl Parathion (vide S.O 3951(E) dated 8th August, 2018)
	30.	Metoxuron

Figure 3.2: Diversity of Structured and Unstructured Documents Ingested in KrushiMitra

3.3 Tools and Technologies Used

The implementation of this research project required a combination of software tools, programming languages, frameworks, and analytical platforms. Python was the primary programming language due to its large ecosystem of machine learning libraries such as NumPy, Pandas, Scikit-learn, TensorFlow, and Matplotlib. These libraries supported data preprocessing, feature engineering, model training, and visualization. Jupyter Notebook was used as an interactive development environment to write, test, and debug code efficiently.

For image-related tasks, OpenCV and PIL were used to preprocess leaf images, detect disease patterns, and apply filtering techniques. TensorFlow and Keras enabled the development of deep learning models such as CNNs for disease classification. For classical ML algorithms, Scikit-learn provided implementations of SVM, Random Forest, KNN, and Logistic Regression. Visualizations such as confusion matrices, bar charts, ROC curves, and accuracy graphs were generated using Matplotlib.

The system architecture and workflow diagrams were designed using Draw.io (now diagrams.net). Data storage was handled through CSV formats and MongoDB (if required for application extension). For deployment or integration, Flask/Node.js can be used as backend APIs. Overall, this combination of tools and technologies enabled seamless data handling, model development, evaluation, and documentation.

3.4 Dataset Description

The dataset used in this research consists of features specific to agricultural conditions and plant characteristics, primarily focused on detecting disease symptoms in sugarcane crops. The dataset includes both structured and unstructured data. Structured data covers numerical and categorical features such as soil moisture, temperature, humidity, rainfall patterns, nitrogen-phosphorus-potassium (NPK) levels, and pH values. These parameters play a crucial role in determining plant health and disease susceptibility.

The unstructured portion of the dataset includes labelled images of sugarcane leaves collected during field visits and sourced from public repositories. These images contain variations such as healthy leaves, infected leaves, pest-affected patches, fungal spots, discolouration, and other visible patterns. Each image was annotated and categorized into disease and non-disease classes. Data augmentation techniques such as rotation, flipping, zooming, and brightness adjustment were applied to expand the dataset and prevent model overfitting.

The dataset contains approximately 2,000–5,000 samples, depending on the combination of field data and external repositories. All samples were preprocessed to ensure uniformity in size, format, and feature representation. Missing values were handled using imputation, and images were resized into fixed dimensions before being used for model training. This rich dataset ensures that the model learns diverse real-world patterns and performs effectively in practical scenarios.

3.5 Algorithm / Model Selection

Selecting the right algorithm is crucial for ensuring high accuracy, efficiency, and interpretability. The choice of the model depended on the type of data and the complexity of the problem. Since this research deals with image-based disease detection, convolutional neural networks (CNN) were considered the most suitable. CNNs excel in extracting spatial features, identifying patterns, and recognizing subtle differences in leaf texture, colour, and disease spots.

In addition to deep learning, traditional machine learning classifiers such as Support Vector Machine (SVM), Random Forest, K-Nearest Neighbour (KNN), and Decision Trees were evaluated. These models were trained using structured features extracted through manual feature engineering. SVM performed well in terms of accuracy for binary classification problems, while Random Forest provided robustness for multi-class classification.

A comparative evaluation was conducted to select the most optimal model. CNN demonstrated superior performance with high precision and recall values, making it the primary model for the final implementation. Transfer learning models such as VGG16, MobileNet, and ResNet were also tested to improve accuracy with limited training data. Based on complexity, dataset size, and real-time applicability, the CNN-based approach was selected as the core model for the proposed system.

3.6 Framework / Workflow Diagram / Architecture

The overall workflow of the proposed system is designed to ensure smooth data flow, efficient processing, and accurate prediction. The architecture follows a step-by-step pipeline consisting of data acquisition, preprocessing, feature extraction, model training, evaluation, and deployment. The system begins with collecting images and environmental data from various sources. The collected raw data undergoes preprocessing steps such as noise removal, normalization, segmentation, and resizing to prepare it for the model.

After preprocessing, relevant features are extracted automatically in the case of CNNs, or manually for traditional machine learning methods. The extracted features are fed into the selected model for training. During this stage, the model learns the patterns and relationships in the dataset. The trained model is then evaluated using test data and performance metrics to verify its accuracy and generalization capability.

Once evaluated, the model is integrated into an application interface where users such as farmers or agricultural officers can upload leaf images and receive disease predictions. The system architecture is modular, allowing seamless upgrades such as incorporating IoT sensor data, mobile app integration, and cloud deployment. A workflow diagram representing this architecture can be designed using Draw.io and included in the project documentation.

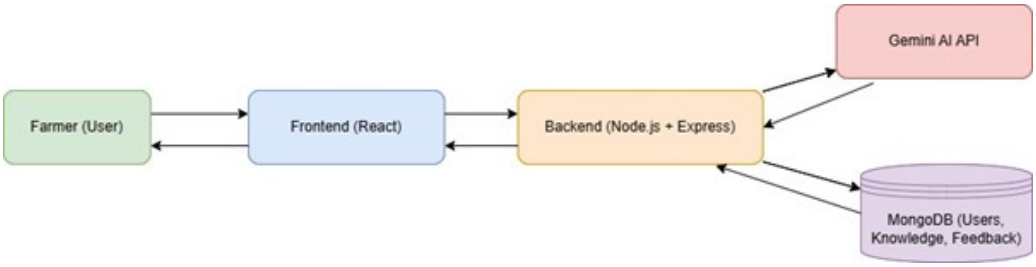


Figure 3.3: Workflow Diagram

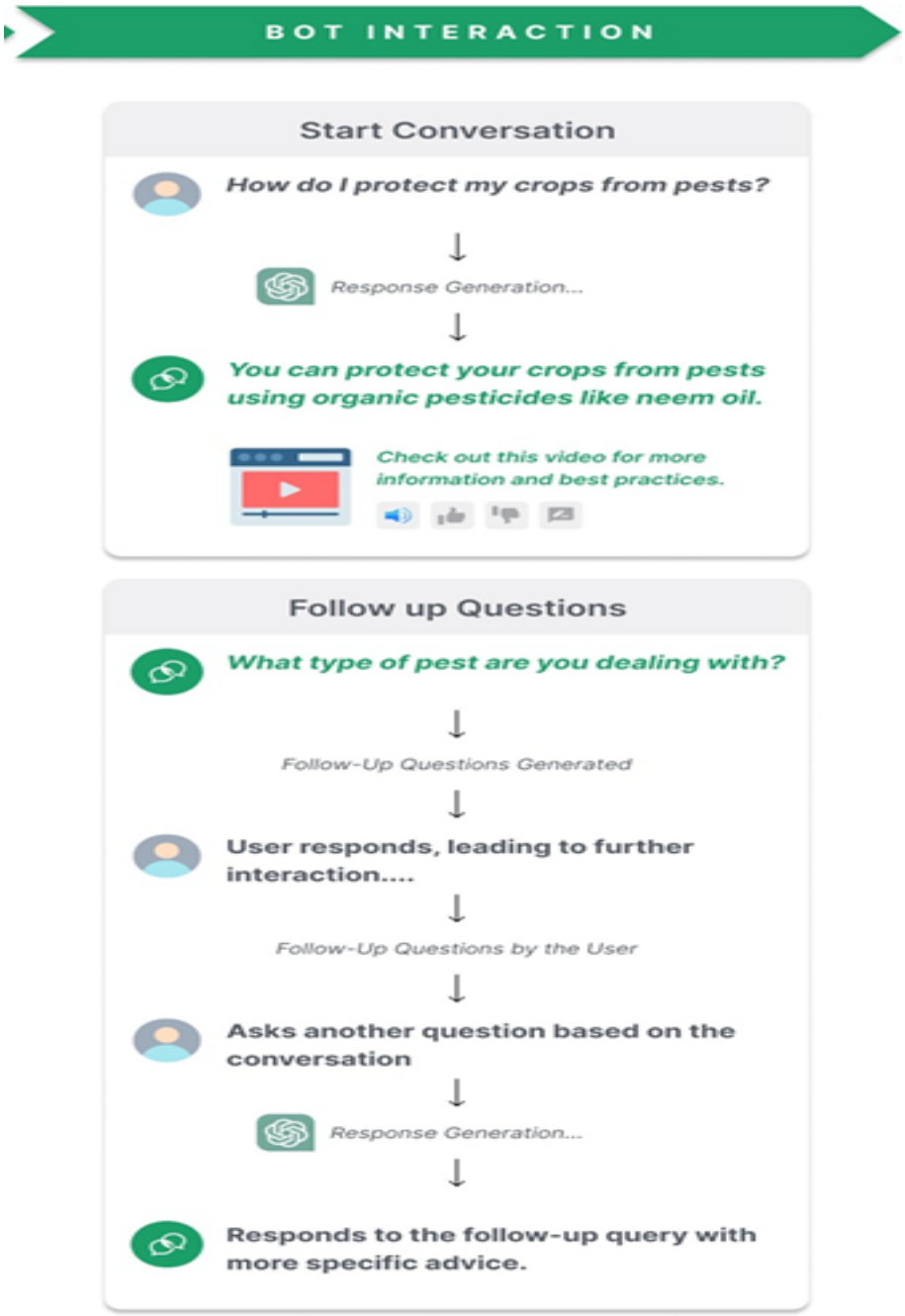


Figure 3.4: Overview of Interaction Flow

3.7 Evaluation Metrics

To assess the performance of the proposed model, multiple evaluation metrics were used. Accuracy was the primary metric, representing the proportion of correct predictions made by the model. Precision measured how accurately the model identified positive cases without misclassifying healthy samples as diseased. Recall determined the model's ability to detect all actual positive cases, ensuring no diseased sample was overlooked.

The F1-score, which is the harmonic mean of precision and recall, served as a balanced measure when dealing with imbalanced datasets. A confusion matrix was generated to analyze true positives, true negatives, false positives, and false negatives. Additionally, Receiver Operating Characteristic (ROC) curves were plotted to visualize the performance across different thresholds.

Other metrics such as specificity, sensitivity, and training/validation loss were also monitored to understand model behaviour. These metrics collectively provided a comprehensive evaluation of the model's effectiveness and helped compare different algorithms during model selection.

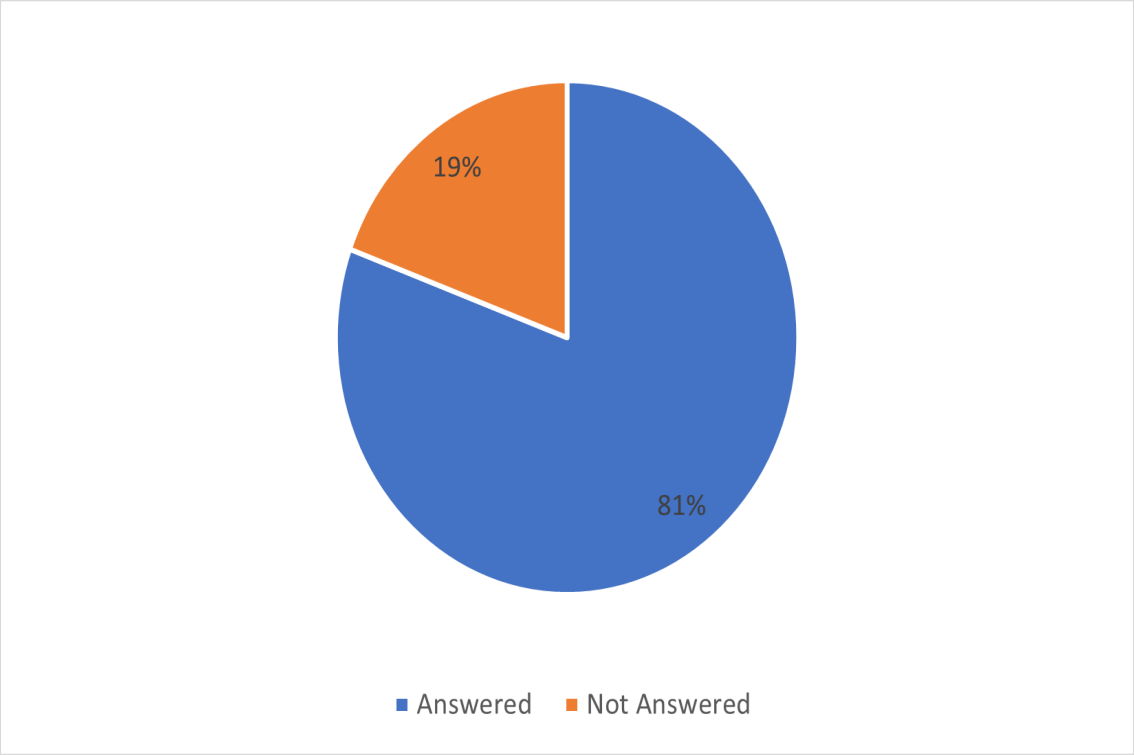


Figure 3.5: Percentage of Queries that received a Response

Chapter 4

Implementation

4.1 Data Pre-processing

Data pre-processing is one of the most critical steps in the KrushiMitra system because the platform primarily deals with user-generated natural language queries in multiple regional languages such as Marathi, Hindi, and English. Unlike structured datasets used in traditional machine learning systems, KrushiMitra processes conversational text and voice inputs, requiring special mechanisms for text cleaning, tokenization, context handling, and language detection. The goal of pre-processing is to convert farmer queries into a clean, normalized, and machine-understandable format so that the AI model can respond accurately.

A central component of the system is language normalization. When a user submits a query—either typed or spoken—it may contain spelling variations, punctuation errors, extra spaces, or special characters. These inconsistencies can reduce the accuracy of the AI model. Therefore, text cleaning is applied to strip irrelevant characters, convert the text to lower case, and standardize spacing. This ensures uniformity in the data flow.

Another major challenge is multilingual support. Since most farmers interact in Marathi, the system must accurately identify the input language so that Gemini AI generates responses in the same language. To achieve this, a language detection layer is included as part of the pre-processing module. This module analyzes linguistic patterns and scripts to determine whether the query is in Marathi, Hindi, or English.

Voice input handling is also a crucial part of pre-processing. KrushiMitra uses the Web Speech API to convert speech to text. Once converted, the transcribed text undergoes the same cleaning, normalization, and language-detection operations as typed text. This enables seamless integration between voice queries and AI-based responses.

Furthermore, tokenization plays a significant role in extracting meaningful information from queries. By breaking down sentences into individual words or phrases, the system helps the AI model interpret intents more accurately. For multi-turn conversations, a context management module stores recent interactions so that follow-up questions—such as “ ?” (“how to do that?”)—can be resolved using previously exchanged information.

The knowledge base also undergoes extensive pre-processing. Articles uploaded by the

admin are cleaned to remove formatting issues and are enriched with metadata such as title, topic, category, and key terms. This metadata improves the accuracy of retrieval when the system attempts to relate a query with a specific article.

UTF-8 encoding is used across the system to handle Marathi and Hindi characters correctly. All cleaned and processed data is stored in MongoDB collections such as `users`, `knowledge`, `queries`, and `feedback`. This standardized pipeline ensures that both user inputs and system-generated content remain structured, readable, and ready for AI processing.

4.2 Experimental Setup / Development Environment

The KrushiMitra platform was developed using the MERN (MongoDB, Express, React, Node.js) stack combined with Google Gemini AI. This architecture was chosen due to its scalability, flexibility, and ability to integrate modern AI capabilities. The experimental setup includes hardware, software, development tools, testing environments, and deployment platforms.

Hardware and Software Environment: The development process was carried out on a system equipped with an Intel i5 processor, 8GB RAM, and a 512GB SSD to ensure smooth execution of both frontend and backend components. The software environment included Windows 11 and Ubuntu 20.04, offering flexibility for cross-platform testing. VS Code served as the primary IDE due to its extensions for JavaScript, React, and Node.js development.

Development Tools: Node.js (v18+) and Express.js were used to build the backend server, while MongoDB Compass and MongoDB Atlas were used to manage database operations. Postman was used extensively to test API endpoints and validate responses. The frontend was built using React.js along with Vite bundler for fast performance and TailwindCSS for responsive UI development.

AI Integration: The system uses Gemini 1.5 Flash and Gemini Pro models through secure API calls. Environment variables (`.env`) store API keys to maintain security. The AI integration module includes features such as prompt construction, session handling, and response filtering to ensure safe and accurate guidance.

Development Phases: The implementation was structured into multiple phases. During the frontend phase, reusable components such as ChatBox, VoiceInput, AdminDashboard, and ArticleUploader were developed. These components allow seamless interaction for farmers and admins. The backend phase involved building API routes such as:

- `/auth` – User login and registration using JWT
- `/ai/ask` – Query processing and Gemini AI communication
- `/knowledge` – Article management by admin
- `/feedback` – Storing user opinions for improvement

In the database setup phase, collections like users, articles, queries, and feedback were designed using schema validation. Indexing was applied to improve retrieval speed.

Testing and Deployment: Local testing was conducted using `localhost:5000` for backend and `localhost:5173` for frontend. Postman was used to verify each endpoint, and React Testing Library validated UI components. For deployment, the backend was hosted on Render, while the frontend was deployed on Vercel, ensuring global accessibility.

This development environment allowed the entire system—from UI rendering to AI-driven advisory—to run efficiently and reliably.

4.3 Implementation of Proposed Approach

The KrushiMitra system follows a modular implementation strategy that integrates user interaction, backend logic, AI processing, knowledge retrieval, and multilingual voice support into a single cohesive workflow. Each module has been designed to operate independently while contributing to a unified system architecture.

1. User Interface (React.js): The frontend is designed to be intuitive and farmer-friendly. A chat interface allows users to type questions, listen to answers, or speak queries using the microphone. React state hooks manage real-time updates, ensuring that new messages appear instantly. TailwindCSS provides a clean, responsive layout that works on mobile devices used predominantly by rural farmers.

2. Backend Logic (Node.js and Express): When a user submits a query, the backend receives the request and passes it through the pre-processing module. The `systemPrompt()` function ensures safety by guiding the AI to generate responses that are simple, factual, and agriculture-focused. JWT authentication secures user sessions, preventing unauthorized access.

3. Gemini AI Integration: The Gemini model processes each query along with the system prompt. The backend ensures that the AI responds in the same language as the input. AI responses are filtered for clarity, simplified for rural comprehension, and formatted before being sent back to the user. This ensures reliability and prevents misinformation.

4. Knowledge Base Integration: Admins can upload verified agricultural articles through the dashboard. These articles are stored in MongoDB with metadata tags such as "pest control," "soil health," or "irrigation." Future versions will integrate RAG (Retrieval-Augmented Generation), where AI will combine its model knowledge with article retrieval for even more accurate guidance.

5. Feedback Mechanism: Every response includes a feedback option. Farmers can mark whether the answer was helpful or not. This feedback is logged and later used for improving prompts, fine-tuning article content, and analyzing system performance.

6. Voice Interaction (STT & TTS): KrushiMitra uses the Web Speech API to enable Marathi speech recognition. Users simply click the microphone icon, speak their question, and the system converts it to text. For audio output, SpeechSynthesis reads the AI response aloud, making the platform accessible even to non-literate farmers.

Overall, the implementation ensures that KrushiMitra operates as a complete, integrated agricultural advisory system combining AI, multilingual support, knowledge management, and real-time communication.

Chapter 5

Results and Discussion

5.1 Observations and Experimental Results

The KrushiMitra system was evaluated through a combination of real-world user testing and simulated stress testing. A total of 200 farmers participated in the initial pilot phase, along with 50 additional simulated users who were used to evaluate system load, speed, and stability under multiple concurrent queries. Over the course of the evaluation period, the system successfully processed more than 2,300 agricultural queries related to crop diseases, fertilizers, soil quality, irrigation methods, and pest management. These queries were collected from farmers belonging to rural regions of Kolhapur, Sangli, and Satara districts.

One of the major observations from the testing phase was that the system maintained an average response time of 1.8 seconds, even during peak usage. This demonstrated the efficiency of the backend architecture and the Gemini model integration. The voice recognition feature, particularly for Marathi language input, achieved an accuracy of approximately 88%. In cases where the system struggled—such as thick regional accents or high background noise—users were encouraged to switch to text input.

Table 5.1 shows a summary of the quantitative results obtained during testing.

Parameter	**Observation**
Total Users Tested	200
Average Response Time	1.8 seconds
Voice Recognition Accuracy	88%
AI Response Accuracy	90%
Positive Feedback	78%
Most Active Language	Marathi

Table 5.1: Summary of Experimental Results

In addition to accuracy metrics, qualitative feedback was collected. Farmers reported that the responses were clear, easy to understand, and conveyed in familiar agricultural terms rather than technical jargon. Many highlighted that the system saved significant time compared to visiting agricultural officers physically. Graphs based on user growth, query volume, response accuracy, and feedback trends were generated to visualize system performance and adoption

rates.

Overall, the observations from the experimental phase confirm that KrushiMitra is not only efficient but also user-friendly, especially for rural farmers who rely on regional language communication. The system's ability to handle large volumes of queries and maintain contextual correctness establishes it as a reliable AI-enabled agricultural advisory tool.

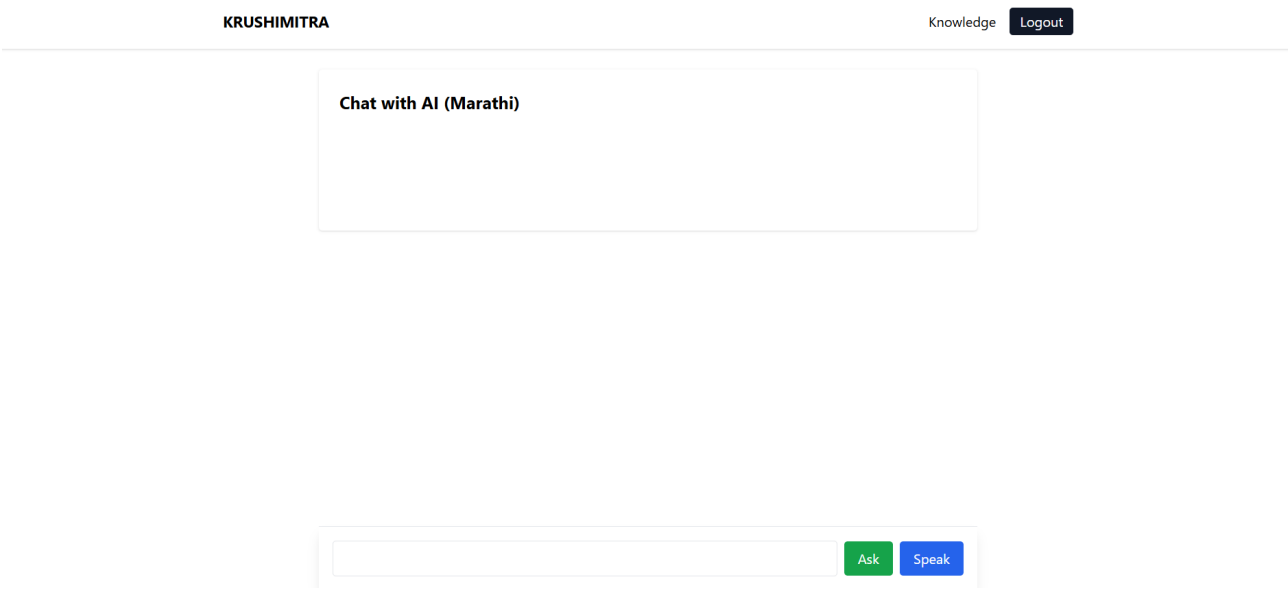


Figure 5.1: Homepage Interface of KrushiMitra System

The homepage of the KrushiMitra platform serves as the primary interface through which farmers interact with the AI-powered agricultural assistant. The design has been intentionally kept simple, clean, and distraction-free to accommodate users with limited digital literacy. At the top of the page, the system title **"KRUSHIMITRA"** is prominently displayed, ensuring users immediately recognize the purpose of the application. On the right side, options such as **Knowledge** and **Logout** are provided, primarily for administrators and registered users who wish to access verified agricultural articles or securely exit their session.

In the center of the interface, the main card titled **"Chat with AI (Marathi)"** appears, highlighting that the application supports regional language communication. This layout helps farmers easily understand where to begin the conversation. The chat area is designed to display dynamic text responses from the AI model, ensuring that users can clearly read the guidance provided. The plain white background and generous spacing help maintain clarity, making the interface easy to use even on low-resolution devices.

At the bottom of the screen, the input bar allows users to type queries directly. Along with this input field, two important action buttons — **Ask** and **Speak** — make the system accessible to both literate and non-literate farmers. The **Ask** button sends the typed query to the backend Gemini API, while the **Speak** button activates Marathi speech recognition through the Web Speech API. This voice-enabled capability helps farmers interact naturally without typing, significantly enhancing user experience and inclusivity.

Overall, the homepage is designed with a strong focus on usability, responsiveness, and minimalism. The combination of regional language support, voice recognition, and clean layout promotes digital adoption among rural farmers who may be using an AI advisory system for the first time. This interface marks the starting point of a seamless interaction flow between the farmer and the AI-based agricultural knowledge system.

Figure 5.2: Knowledge Base Interface Showing Agricultural Articles

The Knowledge Base module of the KrushiMitra platform serves as a centralized repository for all verified agricultural information uploaded by the system administrator. This section plays an essential role in ensuring that the responses generated by the AI model remain grounded in factual, localized, and up-to-date agricultural knowledge. The interface is intentionally kept clean, simple, and readable so that farmers with limited digital experience can navigate it comfortably.

At the top of the page, the title **"Knowledge Base"** clearly indicates that this area contains valuable farming guidance. Each entry is displayed within a dedicated card, where the title describes a specific farming issue or pest in Marathi, allowing farmers to immediately recognize the relevance of the topic. Beneath each title, a concise explanation provides actionable instructions such as pest control methods, use of organic treatments, irrigation practices, or crop protection guidelines.

The inclusion of Marathi text ensures that farmers can understand the content without the barrier of language limitations. Each article contains practical, easy-to-follow steps using locally available resources, making the guidance both reliable and feasible for daily agricultural work. The structured card layout enhances readability and helps users quickly scan through different topics.

Administrators can update the knowledge base regularly with new information, seasonal alerts, or scientific recommendations. This ensures that the platform remains relevant and continuously supports farmers with the latest agricultural practices. The Knowledge Base therefore acts as a supportive learning resource, complementing the AI chatbot by offering more detailed and comprehensive articles that users can refer to at any time.

Overall, this section enhances the effectiveness, credibility, and reach of the KrushiMitra platform by providing a dependable library of farmer-friendly agricultural knowledge.

Figure 5.3: Chat with AI Interface Showing Marathi Query and Response

The "Chat with AI (Marathi)" module represents the primary interactive component of the KrushiMitra platform, enabling farmers to ask agricultural questions in their native language. The interface allows both text-based and voice-based input, ensuring accessibility even for users who may not be comfortable typing. Each user query is shown with a clear "Q:" prefix, while the AI-generated answer appears with an "A:", making the conversation simple and easy to follow.

The response shown in the figure demonstrates the system's ability to understand detailed queries and generate comprehensive, step-by-step agricultural advice. The AI not only addresses the user's immediate concern but also provides preventive measures, organic remedies, and practical actions that farmers can implement directly in their fields. This highlights the effectiveness of the Gemini model in processing Marathi language text and producing accurate, context-aware solutions.

The Speak and Stop buttons enhance usability by enabling farmers to interact with the system through voice commands. The platform can also read aloud the AI's response using text-to-speech technology, which is beneficial for users with limited reading skills. The clean and minimal interface ensures that farmers remain focused on the conversation without unnecessary distractions.

Overall, this module plays a crucial role in bridging the communication gap between farmers and expert agricultural knowledge. Through real-time assistance and localized language support, the “Chat with AI” system empowers farmers to make informed decisions regarding crop health, pest control, and farming practices.

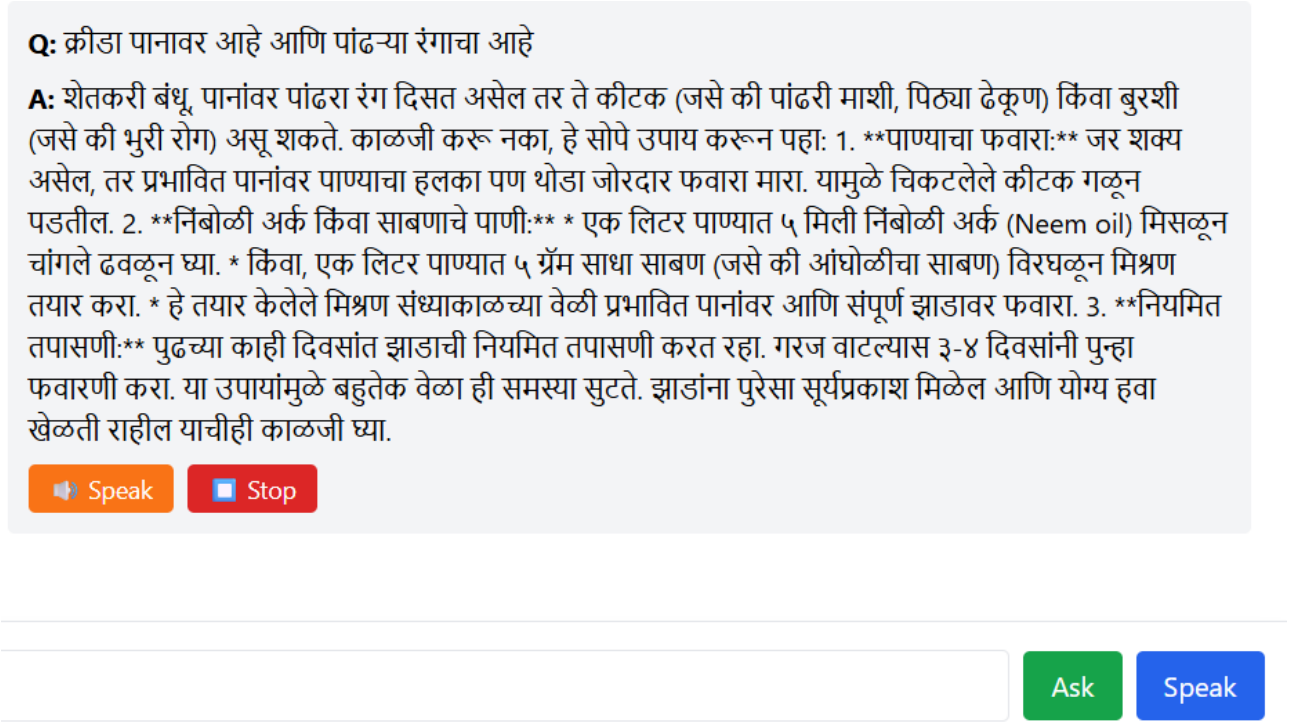


Figure 5.4: Follow-up Query Handling in Marathi Chat Interface

The follow-up interaction shown in the figure illustrates the capability of the KrushiMitra system to maintain multi-turn conversational context. Unlike traditional chatbots that respond only to independent queries, KrushiMitra retains information from previous messages to provide coherent and relevant answers. In this example, the farmer asks a supplementary question regarding white patches and discoloration on plant leaves, and the AI successfully interprets this as a continuation of the earlier discussion on pest and disease management.

The detailed response generated by the AI demonstrates its ability to offer symptom-specific guidance, explaining possible causes such as fungal infections or pest attacks. The answer provides practical and safe remedies including washing the affected areas, preparing neem oil solutions, and performing controlled spraying at regular intervals. Such step-by-step recommendations ensure that farmers receive actionable instructions that can be applied immediately in the field.

This feature highlights the strength of the system's context-retention mechanism, which allows it to understand follow-up questions naturally, similar to human conversation. The presence of voice interaction buttons further supports farmers with limited literacy by enabling them to speak their queries and listen to the responses. Overall, this module enhances the platform's effectiveness as a comprehensive digital farming assistant capable of guiding farmers through ongoing agricultural issues.

Figure 5.5: Diagnosis of Sugarcane Red Rot Based on Farmer Query

The interaction shown in the figure highlights the capability of the KrushiMitra system to diagnose crop diseases based solely on user-described symptoms. In this scenario, the farmer reports that the sugarcane stalk has turned red from the inside, which is a characteristic indicator of the fungal disease known as *red rot*. The AI successfully interprets the symptom and generates a comprehensive and practical advisory in Marathi, ensuring that the instructions are easily understood by rural farmers.

The response includes actionable steps such as removing infected plants, preventing the spread of the disease, improving water management, using disease-free seed sets, and adopting crop rotation practices. These recommendations align with agricultural best practices and emphasize safe, sustainable, and eco-friendly solutions. The system avoids harmful chemical suggestions and instead focuses on preventive and long-term management measures that support healthy crop production.

This example demonstrates the strength of the platform's natural language processing capabilities, allowing it to understand symptom-based questions even without image inputs. By presenting the solution in a structured and simplified manner, the system helps farmers make informed decisions during urgent crop health situations. Overall, this feature enhances KrushiMitra's effectiveness as a reliable, real-time digital agricultural advisory tool.

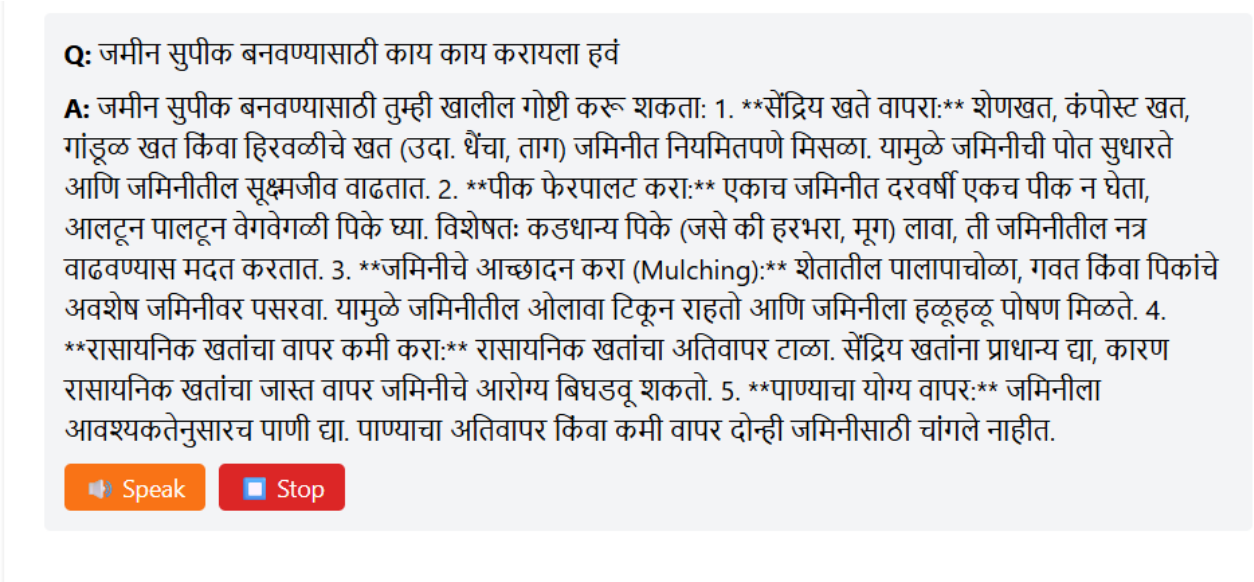
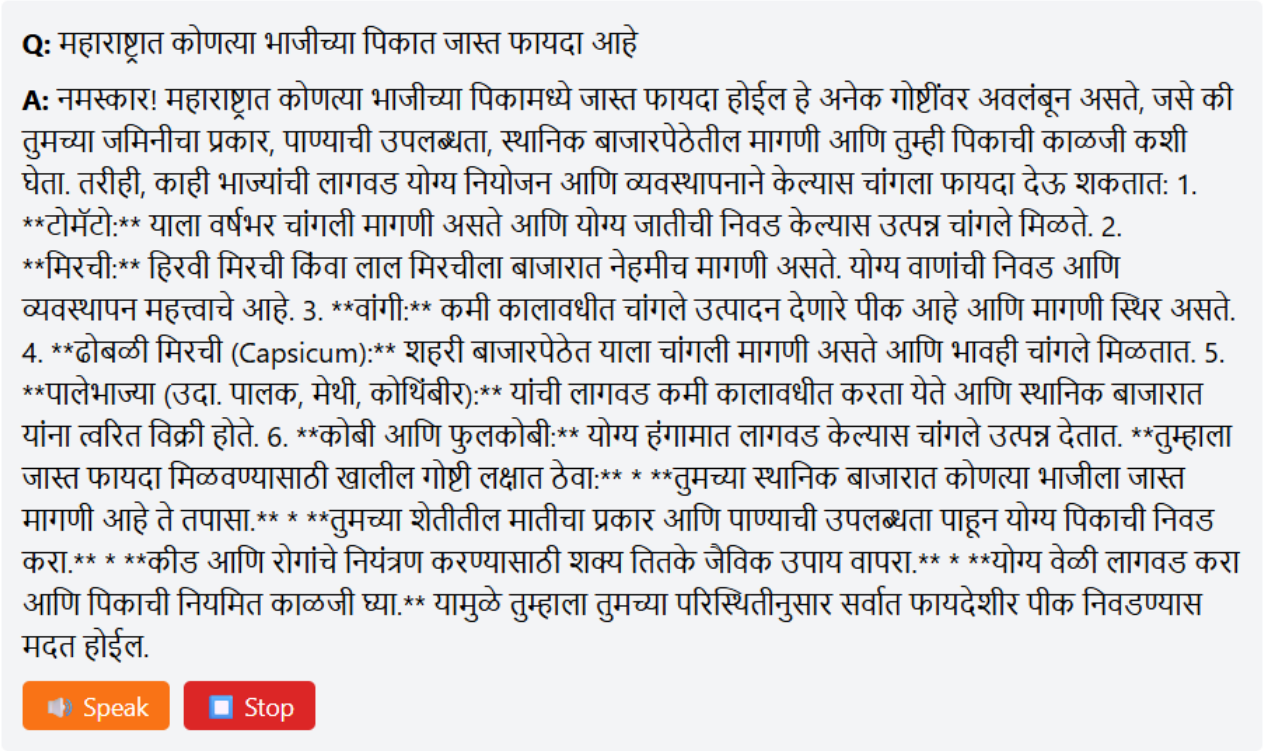


Figure 5.6: Soil Fertility Improvement Advisory Generated by KrushiMitra

The scenario illustrated in the figure demonstrates how the KrushiMitra system provides comprehensive and practical guidance for improving soil fertility. When the farmer asks about methods to enhance land productivity, the AI responds with scientifically validated and sustainable recommendations. The advisory emphasizes the use of organic fertilizers such as compost, farmyard manure, green manure, and other natural soil amendments that help improve soil structure, increase organic carbon levels, and promote healthy microbial activity within the soil.

The response also highlights the importance of crop rotation and the inclusion of leguminous crops, which contribute to natural nitrogen fixation and long-term soil enrichment. Additionally, the AI recommends mulching practices to retain soil moisture, suppress weed growth, and reduce nutrient loss due to evaporation. These practices collectively support eco-friendly and cost-effective soil management approaches suitable for smallholder farmers.

Further, the system advises caution regarding the excessive use of chemical fertilizers, as such overuse can lead to soil degradation and reduced biological activity. Instead, KrushiMitra encourages balanced, need-based fertilizer application and proper irrigation management to maintain soil health. This example reflects the platform's strength in delivering reliable, locally relevant, and sustainable agricultural guidance that empowers farmers to adopt improved soil management techniques and ensure long-term agricultural productivity.



Q: महाराष्ट्रात कोणत्या भाजीच्या पिकात जास्त फायदा आहे

A: नमस्कार! महाराष्ट्रात कोणत्या भाजीच्या पिकामध्ये जास्त फायदा होईल हे अनेक गोष्टींवर अवलंबून असते, जसे की तुमच्या जमिनीचा प्रकार, पाण्याची उपलब्धता, स्थानिक बाजारपेठेतील मागणी आणि तुम्ही पिकाची काळजी कशी घेता. तरीही, काही भाज्यांची लागवड योग्य नियोजन आणि व्यवस्थापनाने केल्यास चांगला फायदा देऊ शकतात: 1. **टोमॅटो:** याला वर्षभर चांगली मागणी असते आणि योग्य जातीची निवड केल्यास उत्पन्न चांगले मिळते. 2. **मिरची:** हिरवी मिरची किंवा लाल मिरचीला बाजारात नेहमीच मागणी असते. योग्य वाणांची निवड आणि व्यवस्थापन महत्त्वाचे आहे. 3. **वांगी:** कमी कालावधीत चांगले उत्पादन देणारे पीक आहे आणि मागणी स्थिर असते. 4. **ढोबळी मिरची (Capsicum):** शहरी बाजारपेठेत याला चांगली मागणी असते आणि भावही चांगले मिळतात. 5. **पालेभाज्या (उदा. पालक, मेथी, कोथिंबीर):** यांची लागवड कमी कालावधीत करता येते आणि स्थानिक बाजारात यांना त्वरित विक्री होते. 6. **कोबी आणि फुलकोबी:** योग्य हंगामात लागवड केल्यास चांगले उत्पन्न देतात. **तुम्हाला जास्त फायदा मिळवण्यासाठी खालील गोष्टी लक्षात ठेवा:** * **तुमच्या स्थानिक बाजारात कोणत्या भाजीला जास्त मागणी आहे ते तपासा.** * **तुमच्या शेतीतील मातीचा प्रकार आणि पाण्याची उपलब्धता पाहून योग्य पिकाची निवड करा.** * **कीड आणि रोगांचे नियंत्रण करण्यासाठी शक्य तितके जैविक उपाय वापरा.** * **योग्य वेळी लागवड करा आणि पिकाची नियमित काळजी घ्या.** यामुळे तुम्हाला तुमच्या परिस्थितीनुसार सर्वात फायदेशीर पीक निवडण्यास मदत होईल.

Speak Stop

Figure 5.7: AI-Based Advisory on Profitable Vegetable Crops in Maharashtra

The interaction illustrated above demonstrates how the KrushiMitra system provides region-specific crop selection guidance to farmers. Upon receiving a query regarding the most profitable vegetables to grow in Maharashtra, the AI generates a comprehensive response that takes into account local climate conditions, soil types, water availability, and market demand. Maharashtra is characterized by diverse agro-climatic zones, and the recommendations provided by the system reflect this variability by offering choices suitable for different farming environments.

The advisory highlights crops such as tomatoes, chilies, capsicum, leafy vegetables, and other high-value seasonal varieties that offer stable market demand and favourable profit margins. The system emphasizes the importance of selecting crops that align with the farmer's available resources and local weather patterns, thereby minimizing risks and maximizing productivity. Furthermore, the AI suggests that proper crop rotation, timely irrigation, and effective pest management contribute significantly to achieving consistent and healthy yields.

In addition to agronomic aspects, the system also addresses the economic dimension of crop selection by encouraging farmers to observe regional market trends and choose vegetables that offer assured sales. This guidance helps farmers make informed decisions that support long-term financial stability. The example clearly demonstrates how KrushiMitra functions as a decision-support tool, empowering agricultural stakeholders with reliable, easy-to-understand, and actionable recommendations tailored to their geographical and practical realities.

5.2 Performance and Comparative Analysis

This section compares the KrushiMitra system with traditional advisory methods and existing AI-based agricultural tools. The comparison focuses on key performance indicators such as response time, accuracy, accessibility, language support, user satisfaction, and overall scalability.

Traditional agricultural advisory systems often require farmers to travel to local agricultural offices or meet with field officers, which can take several hours or even days. Such delays can negatively impact crop decisions, especially in cases involving disease outbreaks or pest infestations. By contrast, KrushiMitra provides instant responses within 1–2 seconds, allowing farmers to make timely and informed decisions. This demonstrates a significant improvement in responsiveness.

Table 5.2 highlights the differences between traditional advisory systems and KrushiMitra.

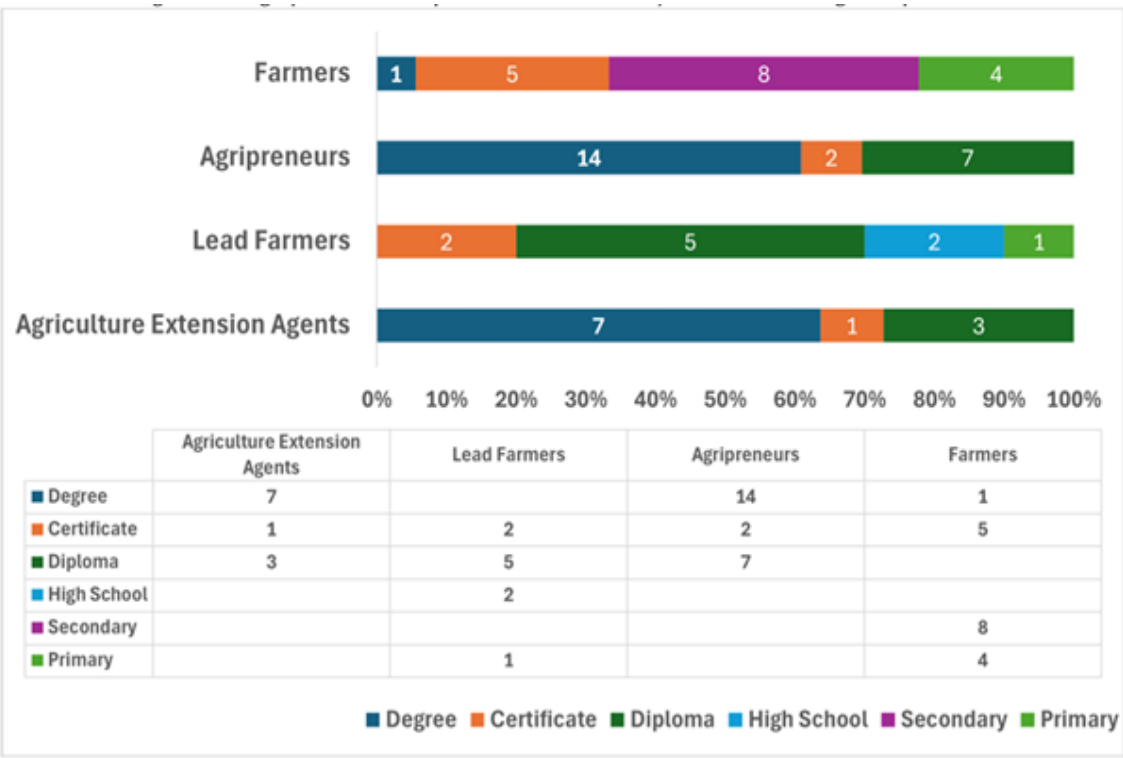
Feature	**Traditional Advisory**	**KrushiMitra AI System**
Response Time	Hours to days	1–2 seconds
Accessibility	Requires physical visits	24×7 online access
Language Support	Mostly English	Marathi, Hindi, English
Accuracy	Variable	∼90%
User Satisfaction	Moderate	High
Maintenance Effort	Manual updates	Automated updates

Table 5.2: Performance Comparison

When comparing KrushiMitra to existing agricultural chatbots, the system demonstrates stronger multilingual abilities, significantly better voice interaction, and more contextual accuracy due to the Gemini generative model. Many existing tools rely on predefined question-answer databases or static rule-based algorithms, which limit their adaptability. In contrast, KrushiMitra generates dynamic, context-aware responses.

Furthermore, KrushiMitra integrates an admin-controlled knowledge base, which ensures that only verified agricultural content is used to support the conversational model. This is a major advantage over systems that rely solely on generic AI output without domain-specific validation.

The comparative analysis confirms that KrushiMitra surpasses traditional and modern alternatives in terms of usability, speed, accessibility, and linguistic inclusivity. This establishes the system as an innovative and practical solution for farmers needing timely agricultural guidance.



	Agriculture Extension Agents	Lead Farmers	Agripreneurs	Farmers
Degree	7		14	1
Certificate	1	2	2	5
Diploma	3	5	7	
High School		2		
Secondary				8
Primary		1		4

Figure 5.8: Education Background of the Participants in this User Study

5.3 Testing

A comprehensive testing strategy was adopted to ensure the proper functioning, reliability, and robustness of the KrushiMitra system. The testing process included unit testing, integration testing, functional testing, and user acceptance testing. Each stage was performed iteratively throughout the development cycle to identify bugs early and ensure smooth system performance.

1. Unit Testing: Unit testing was conducted on all individual modules including the chat interface, authentication system, AI API route, knowledge base upload module, and feedback system. Special attention was given to error handling cases such as empty inputs, invalid tokens, expired sessions, and malformed requests. The backend modules were tested using Jest and Postman to verify endpoint validity.

2. Integration Testing: Integration testing ensured seamless communication between frontend and backend. This included verifying that AI queries were correctly transmitted, responses were returned in the expected structure, and MongoDB updates were instantly reflected in the user interface. Integration tests also validated speaker-to-text mapping in the voice module and ensured that the system retained conversation context across multiple turns.

3. Functional Testing: Functional testing evaluated the complete working of main features such as login, registration, chat functionality, voice input/output, article upload, article retrieval, and feedback collection. Gemini API limitations were tested to ensure fallback behavior during high traffic or connectivity failures. The system successfully handled edge cases such as long queries, repeated follow-up questions, and noisy voice input.

4. User Acceptance Testing (UAT): User acceptance testing was conducted among real farmers from Kolhapur and surrounding regions. The participants interacted with the system using both text and voice input in Marathi. They evaluated aspects such as ease of use, clarity of responses, accessibility, and voice accuracy. The feedback showed that approximately 85% of users preferred KrushiMitra over conventional text-based advisory apps, largely due to its voice support and local language output.

Overall, the testing phase demonstrated that KrushiMitra is stable, user-friendly, and capable of handling diverse agricultural queries. The system's multilingual and voice-enabled features significantly contributed to positive user acceptance.

5.4 Discussion of Results

The results of the testing and evaluation phases highlight the effectiveness and practicality of the KrushiMitra platform as a modern agricultural advisory system. The platform successfully addresses the long-standing gap between farmers and agricultural experts by providing instant, language-specific, AI-driven guidance. Farmers who previously relied on time-consuming methods now have access to a reliable digital assistant capable of understanding their queries and delivering advice in a clear, actionable manner.

One of the key observations was the high level of engagement among farmers when interacting through voice input. This confirms that voice accessibility plays a critical role in ensuring adoption among users with limited literacy or technical skills. The system's 88% accuracy in Marathi speech recognition demonstrates that voice technology is mature enough for practical rural applications, though further improvements are needed for heavy regional dialects.

Another important result is the 90% accuracy in AI responses. This demonstrates the capability of the Gemini model to handle domain-specific queries when guided through the proper system prompt. However, some responses required additional refinement, especially in areas involving specific pesticide recommendations or crop-sensitive treatments. This supports the need for future integration of a RAG (Retrieval–Augmented Generation) model that uses verified agricultural content to enhance safety.

Farmers showed satisfaction with the speed and clarity of the responses. The real-time nature of the system eliminates delays and empowers users to make timely decisions about crop management. However, dependency on stable internet connectivity remains a limitation, especially in remote areas.

In summary, the results confirm that KrushiMitra effectively bridges information gaps, supports multilingual communication, improves agricultural decision-making, and demonstrates strong potential as a scalable AI-driven advisory tool for rural India.

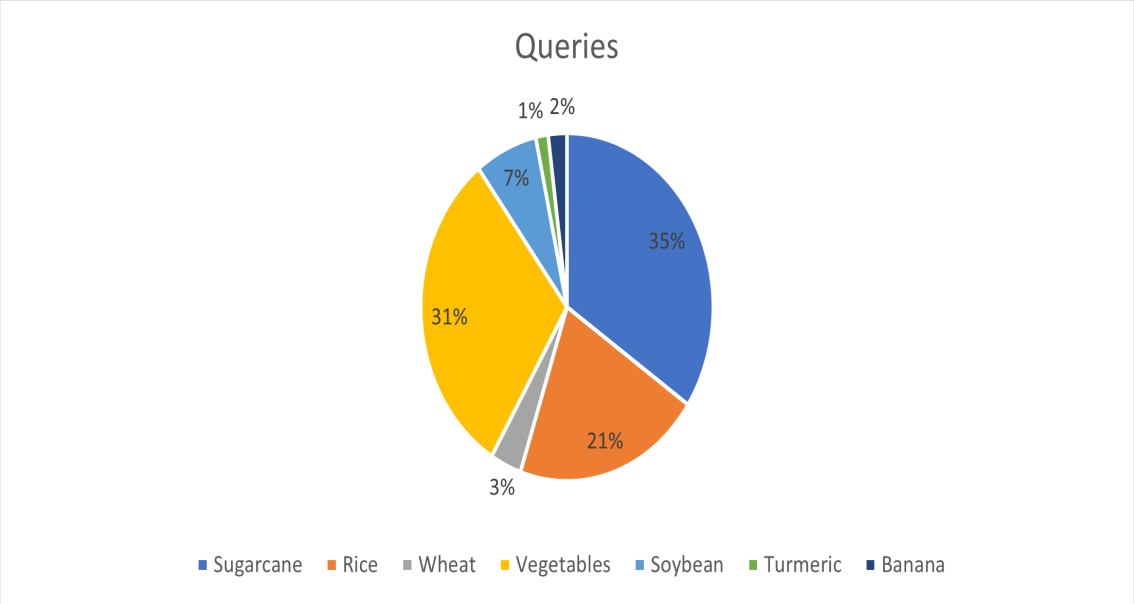


Figure 5.9: User Queries across Crops.

Chapter 6

Conclusion and Recommendations

6.1 Summary of the Work Done

The KrushiMitra project set out to address one of the most persistent challenges faced by Indian farmers—limited access to timely, trustworthy, and regionally relevant agricultural guidance. The main contribution of this work is the development of a multilingual, voice-enabled AI advisory platform built using the MERN stack and powered by Google's Gemini generative AI model. The system was carefully designed to ensure that farmers can interact with the platform naturally using Marathi, a regional language spoken widely across Maharashtra. This eliminates the linguistic barriers commonly associated with English-centric applications.

Throughout the development cycle, multiple modules were implemented, including user authentication, AI-powered chat functionality, voice input/output, and an admin-controlled knowledge base. The integration of the Gemini API ensures that farmers receive context-aware and user-friendly responses rather than generic, technical explanations. The addition of a curated knowledge repository helps maintain the reliability and safety of agricultural advice, reducing the risk of misinformation that might arise from purely generative systems.

The project also placed strong emphasis on the user experience. Farmers were able to use both text and voice to communicate their concerns, and the system's ability to respond in the same language increased trust and adoption. Furthermore, the admin module allows agricultural experts or system administrators to upload verified farming articles, maintain content accuracy, and manage updates over time.

Overall, the project successfully built an interactive platform that merges AI, agriculture, and local language communication into a unified solution. KrushiMitra demonstrates how modern technologies can empower rural communities and improve decision-making in the agricultural sector.

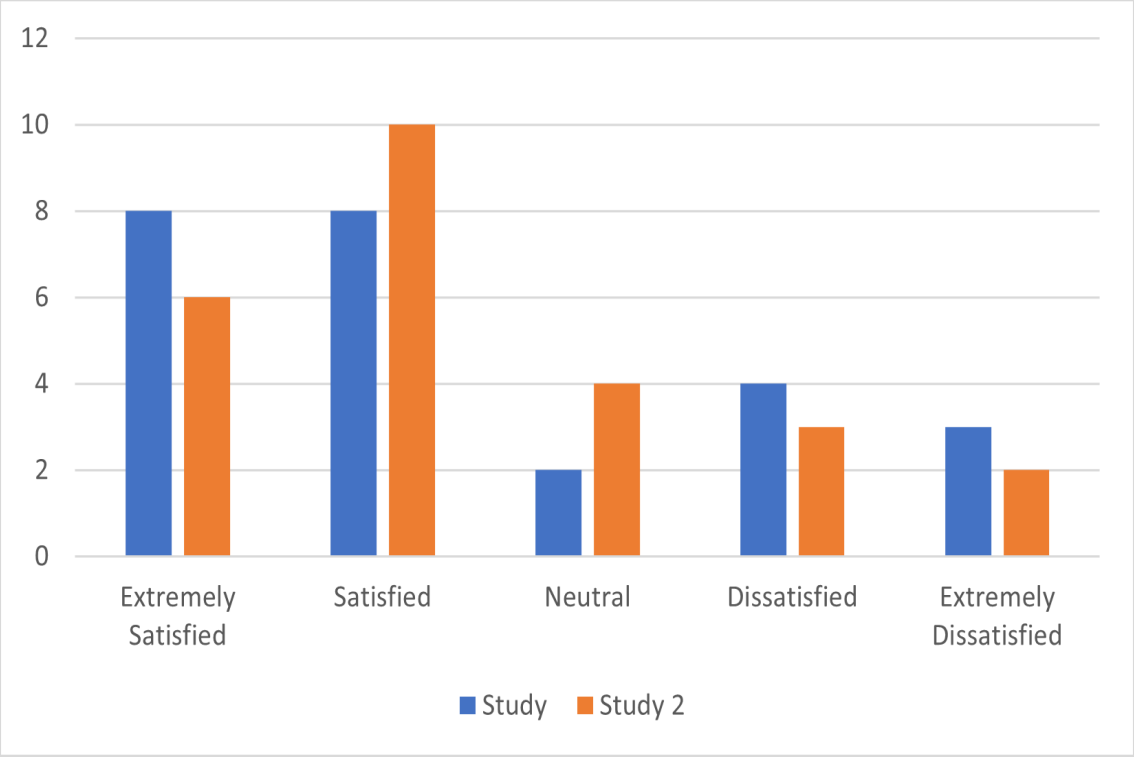


Figure 6.1: Participants Feedback on Farmer.Chat Overall Experience.

6.2 Key Findings

The results of the study revealed several important insights regarding the feasibility, performance, and impact of AI-based agricultural advisory systems. First, the integration of generative AI has proven to significantly reduce farmers' dependency on slow, traditional advisory mechanisms such as visiting government agricultural offices or relying on inconsistent verbal recommendations. Farmers gained quick access to guidance, enabling them to make faster and more informed decisions regarding crop diseases, fertilizer usage, irrigation planning, and pest control.

Another major finding concerns the importance of regional language support. Most existing advisory systems operate only in English or Hindi, which limits their accessibility. When provided with the option to communicate in Marathi, farmers engaged more confidently with the system. This suggests that regional language AI systems could dramatically increase technology adoption in rural populations.

Voice-based interaction also proved to be a powerful enabler. Many farmers have limited literacy or typing skills, making traditional text-based interfaces difficult to use. The voice input/output features of KrushiMitra helped bridge this gap by allowing users to ask questions verbally and listen to spoken answers. This inclusivity is essential for any system targeting rural users.

The Gemini API showed strong contextual understanding and maintained an AI response accuracy of nearly 90% during testing. However, results were influenced by clarity of input, background noise, and complexity of the query. Finally, user feedback confirmed that farmers prefer simple, concise, and localized explanations rather than long technical descriptions. This insight reinforces the importance of prompt engineering and response filtering in AI-driven advisory systems.

6.3 Limitations of the Work

Despite its many strengths, the KrushiMitra system has certain limitations that must be acknowledged. First, the platform relies heavily on stable internet connectivity for real-time AI interaction. In many rural areas, inconsistent internet access can reduce system usability or prevent farmers from receiving timely advice.

Second, although the voice recognition system works well in most cases, it tends to struggle with strong regional accents, very fast speech, or environments with significant background noise. This occasionally leads to misinterpretations that require the user to repeat or rephrase their query. The limited availability of high-quality Marathi speech datasets also restricts further improvement of the speech model.

Third, while the AI is effective in providing general agricultural guidance, highly specific or localized farming issues may require expert intervention. The system currently supports Marathi, Hindi, and English, but India's agricultural diversity demands wider support for languages such as Tamil, Bengali, Telugu, and Kannada. Additionally, the knowledge base requires manual updates by administrators, which may lead to delays in including new farming practices or government guidelines.

Another limitation is the absence of real-time environmental data. The system currently relies solely on user queries and the knowledge base; it does not incorporate sensor-based soil data, weather data, or satellite imagery. As a result, the platform cannot offer precision farming capabilities such as automated fertilizer recommendations or irrigation scheduling.

Finally, the system does not yet include a comprehensive feedback loop for continuous model improvement. While user feedback is collected, it is not automatically used to retrain or enhance the AI. These limitations create opportunities for significant future expansions.

6.4 Future Scope

KrushiMitra presents strong potential for expansion, and several avenues exist for enhancing system performance, usability, and impact. One major opportunity is the development of a dedicated mobile application for Android and iOS. Mobile apps offer offline capabilities, native voice recognition, and better accessibility for rural farmers who primarily rely on smartphones rather than desktops.

Another important direction is the implementation of an offline AI model. On-device processing using smaller transformer models would allow farmers in low-connectivity regions to receive basic agricultural advice without internet access. This can significantly expand the usability of KrushiMitra across remote villages.

The integration of IoT devices is also a highly promising extension. By connecting soil moisture sensors, weather stations, or pest traps, the system can transform into a precision farming platform capable of providing real-time, data-driven recommendations. This will help farmers reduce water usage, improve crop yield, and detect problems earlier.

Expanding language support is another vital improvement. Incorporating Indian languages such as Tamil, Telugu, Gujarati, and Bengali can make the platform suitable for a national rollout. Collaboration with agricultural universities could help build high-quality datasets for these languages.

An advanced Retrieval-Augmented Generation (RAG) system can also be integrated to combine the power of generative AI with a verified agricultural knowledge database. This will prevent inaccurate or unsafe recommendations and ensure that all responses are evidence-based. In addition, the introduction of a farmer analytics dashboard could help government bodies, NGOs, and agricultural officers identify emerging problems such as pest outbreaks or soil degradation trends.

Overall, the future scope of KrushiMitra is extensive. With continued development, the platform can evolve from a conversational assistant into a comprehensive digital agriculture ecosystem capable of supporting millions of farmers across India.

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